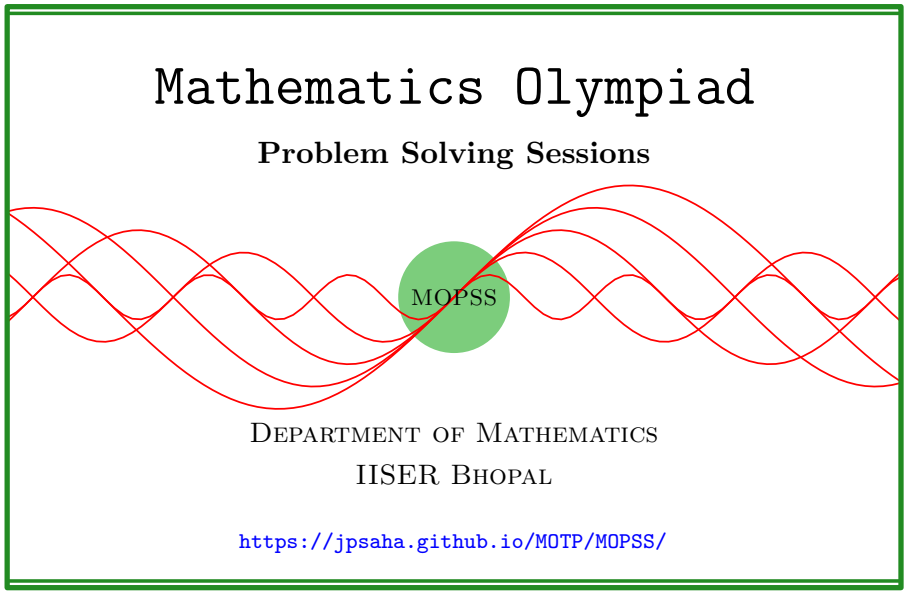
Congruence and similarity



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

1 Exercise (RMO 2007 P5,) 2
2 Exercise (RMO 1998 P4,) 2
3 Exercise (RMO 2012e P1,) 2
4 Exercise (RMO 2003 P1,) 2
5 Exercise (RMO 2013b P4,) 2
6 Exercise (RMO 2013b P2,) 2

§1 Congruence and similarity of triangles

Exercise 1 (RMO 2007 P5,). A trapezium $ABCD$, in which AB is parallel to CD, is inscribed in a circle with centre O. Suppose the diagonals AC and BD of the trapezium intersect at M, and $OM = 2$.

(a) If $\angle AMB$ is $60°$, determine, with proof, the difference between the lengths of the parallel sides.

(b) If $\angle AMD$ is $60°$, find the difference between the lengths of the parallel sides.

Exercise 2 (RMO 1998 P4,). Let ABC be a triangle with $AB = AC$ and $\angle BAC = 30°$. Let A' be the reflection of A in the line BC; B' be the reflection of B in the line CA; C' be the reflection of C in line AB. Show that $A'B'C'$ is an equilateral triangle.

Exercise 3 (RMO 2012e P1,). Let $ABCD$ be a convex quadrilateral such that $\angle ADC = \angle BCD > 90°$. Let E be the point of intersection of AC and the line through B parallel to AD; let F be the point of intersection of BD and the line through A parallel to BC. Prove that $EF \parallel CD$.

Exercise 4 (RMO 2003 P1,). Let ABC be a triangle in which $AB = AC$ and $\angle CAB = 90°$. Suppose M and N are points on the hypotenuse BC such that $BM^2 + CN^2 = MN^2$. Prove that $\angle MAN = 45°$.

Exercise 5 (RMO 2013b P4,). Let ABC be a triangle with $\angle A = 90°$ and $AB = AC$. Let D and E be points on the segment BC such that $BD : DE : EC = 1 : 2 : \sqrt{3}$ $(3 : 4 : 5)$. Prove that $\angle DAE = 45°$.

Exercise 6 (RMO 2013b P2,). In a triangle ABC, AD is the altitude from A, and H is the orthocentre. Let K be the centre of the circle passing through D and tangent to BH at H. Prove that the line DK bisects AC.

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)

The content posted here and at this blog by Evan Chen are quite useful.